Hallucination Guardrails via MFA and EVT

Gella Naga Sai Krishna · Data Science, Semester 5 · November 2026 · Code:
github.com/GellaNagaSaiKrishna/Hallucination-Guardrails-via-MFA-and-EVT

1. Problem and approach

A Retrieval-Augmented Generation (RAG) system retrieves the closest chunks of a knowledge base and asks an LLM to answer from them. Nearest-neighbour retrieval always returns *something*, even when the knowledge base cannot answer the question, and the LLM then often produces a fluent but unsupported answer (a hallucination). This project builds a **pre-retrieval guardrail**: a statistical test on the question's embedding that refuses out-of-distribution (OOD) questions before retrieval and generation. The test combines a **Mixture of Factor Analyzers (MFA)** density, a fast **Woodbury** Mahalanobis distance and an **Extreme Value Theory (EVT)** p-value.

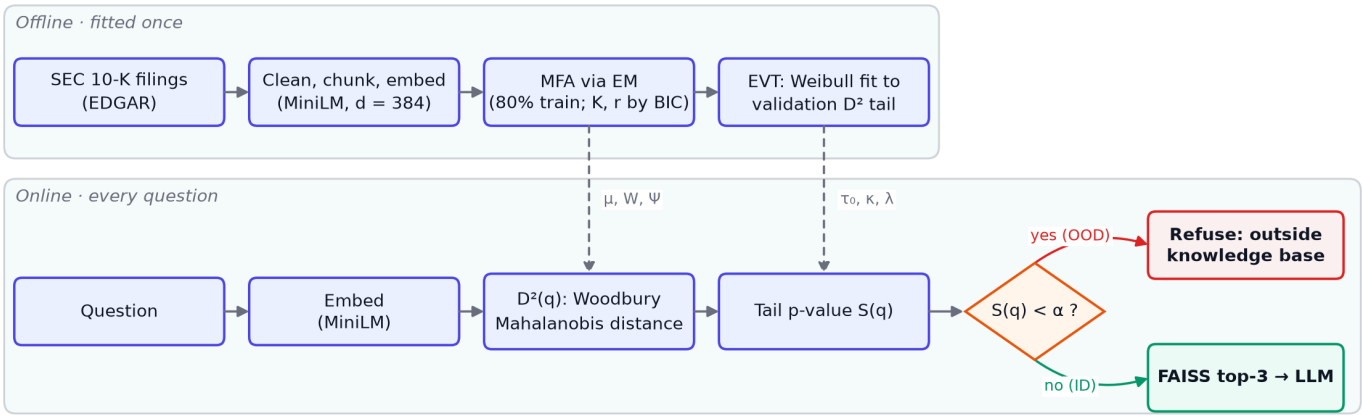


Figure 1. Offline, the MFA is fitted on training chunks and the EVT tail on validation distances. Online, a question's distance is turned into a p-value S(q) and the question is refused if S(q) < α.

2. Data

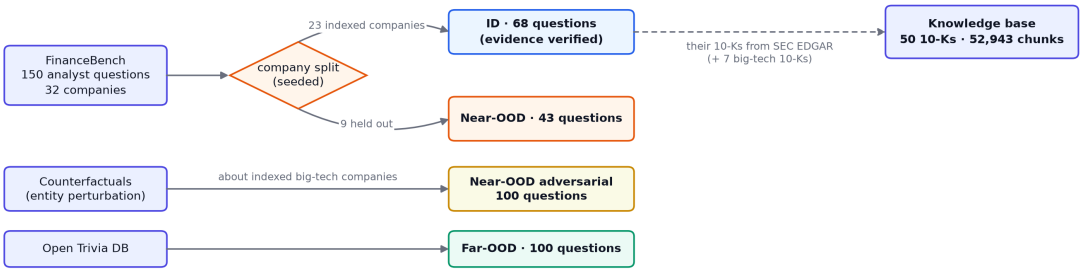


Figure 2. How the knowledge base and the four question sets were built.

Set	Size	Source
Knowledge base	50 10-K filings → 52,943 chunks	SEC EDGAR: the FinanceBench filings of 23 companies + the latest 10-Ks of 7 big-tech companies
ID	68	FinanceBench (Islam et al., 2023), questions whose evidence passage is found in our text (median coverage 97%)
Near-OOD	43	FinanceBench questions on 9 held-out companies (AMD, Adobe, Block, Boeing, General Mills, MGM, Netflix, PayPal, Verizon)
Near-OOD adversarial	100	constructed counterfactuals about indexed companies (e.g. *Apple Car revenue*), a false-premise stress test

Set	Size	Source
Far-OOD	100	Open Trivia Database

Filings are parsed with BeautifulSoup (inline-XBRL header, page numbers and repeated headers removed), split into 500-character chunks with 50 overlap, embedded with all-MiniLM-L6-v2 (d = 384) and split at random into **42,355 training** and **10,588 validation** chunks. Test questions are never used for fitting. Walkthrough: notebook 01.

3. Method

3.1 Mixture of Factor Analyzers · notebook 02

Each of K clusters explains an embedding by r ≪ d latent factors plus independent noise, giving a low-rank-plus-diagonal covariance:

```
p(x) = Σ_k π_k N(x | μ_k, W_k W_k^T + Ψ_k),     W_k ∈ R^(d×r),  Ψ_k diagonal
```

EM is implemented from scratch (src/mfa.py): K-Means + PPCA initialisation, E-step responsibilities and factor posteriors, AECM M-step, Ψ floored at 10^{-4}·Var. No d × d matrix is formed. The **Woodbury identity** gives the distance through an r × r Cholesky factor in O(K·d·r):

```
Σ^-1 = Ψ^-1 − Ψ^-1 W (I + W^T Ψ^-1 W)^-1 W^T Ψ^-1,     D²(q) = min_k (q − μ_k)^T Σ_k^-1 (q − μ_k)
```

K and r were chosen by BIC over K ∈ {2, 4, 8, 16, 32} × r ∈ {8, 16, 32, 64}. **Selected: K = 16, r = 64**, converged in 48 EM iterations (6.7 min), **373,263 parameters** against 1,188,879 for a full-covariance GMM (−69%); cluster weights 3.5–10.1%.

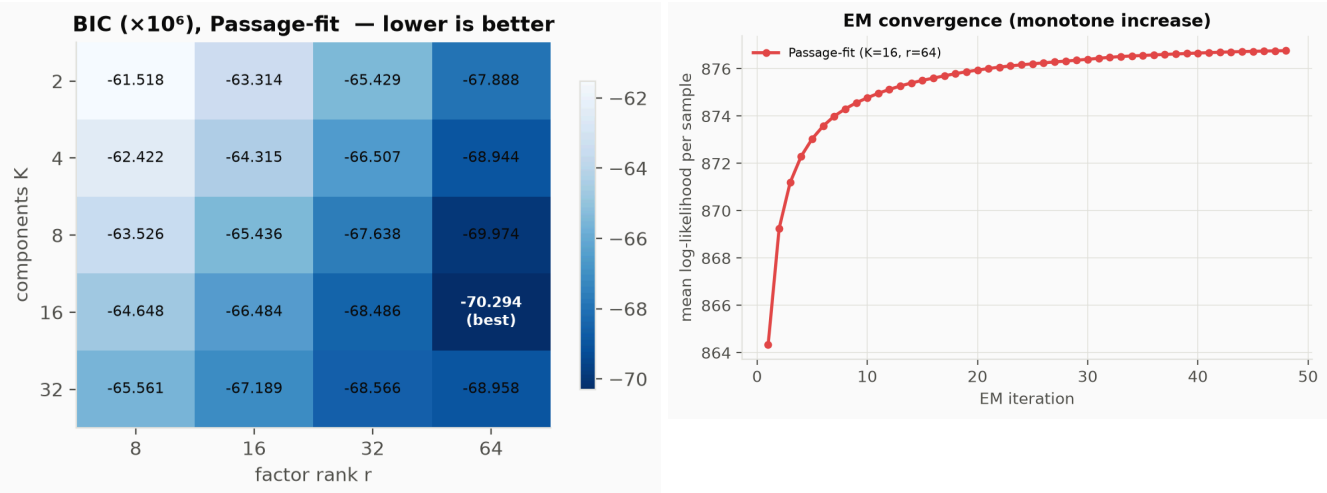


Figure 3. Left: BIC improves with rank in every row and is U-shaped in K, with the optimum at K = 16; at K = 32 the parameter penalty outweighs the gain. r = 64 is the grid edge. Right: EM increases the log-likelihood monotonically and flattens, as theory requires.

3.2 Extreme Value Theory · notebook 03

An empirical 99th percentile rests on very few points. Peaks-over-threshold EVT instead models the whole tail. With τ_0 the 90th percentile of validation D^2, a Weibull distribution is fitted by maximum likelihood to the excesses $D^2 - \tau_0$ (src/evt.py), and a question's p-value is

```
S(q) = exp( −( max(0, D²(q) − τ₀) / λ )^κ ),     reject if S(q) < α
equivalent distance threshold:  D* = τ₀ + λ (−ln α)^(1/κ)
```

Fitted: τ_0 = 483.1, κ = 1.053 (≈ exponential tail), λ = 50.85, from 1,059 exceedances. The fit is not rejected: KS p = 0.57, or p = 0.16 with a parametric bootstrap that accounts for the estimated parameters. **D* = 700** at α = 0.01 and 595 at α = 0.1; moving τ_0 to the 85th or 95th percentile changes D* by only −2% / +3%. EVT does not change how questions are ranked, only where the line is drawn.

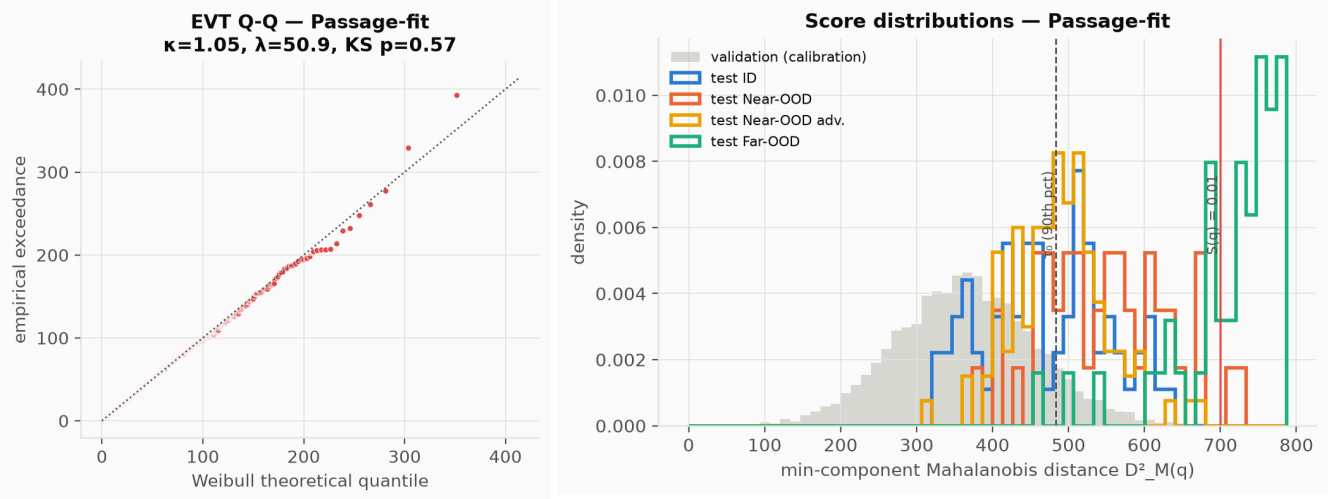


Figure 4. Left: the exceedances lie on the Weibull Q–Q diagonal except for the few most extreme points. Right: median D^2 is 364 for validation chunks, 466 for ID questions, 490 adversarial, 548 Near-OOD and 793 Far-OOD. Far-OOD lies beyond the α = 0.01 threshold; Near-OOD is shifted right but overlaps ID; adversarial questions sit on top of ID. ID questions lie further out than passages (query–passage asymmetry).

4. Results

Baselines are fitted on the same training chunks (K = 16): cosine similarity to the nearest chunk (FAISS), distance to the nearest K-Means centroid, and diagonal and full-covariance GMMs. Full tables: notebook 04.

4.1 Detection

Method	AUROC Near-OOD	AUROC adversarial	AUROC Far-OOD	AUROC all
MFA + EVT (proposed)	**0.736**	0.559	0.988	0.767
MFA log-likelihood	**0.756**	0.608	0.991	0.792
Cosine NN (FAISS)	0.662	**0.762**	**0.996**	**0.841**
K-Means Euclidean	0.481	0.669	**0.996**	0.770
Diagonal GMM	0.507	0.656	0.994	0.769
Full-covariance GMM	**0.772**	0.619	0.986	0.797

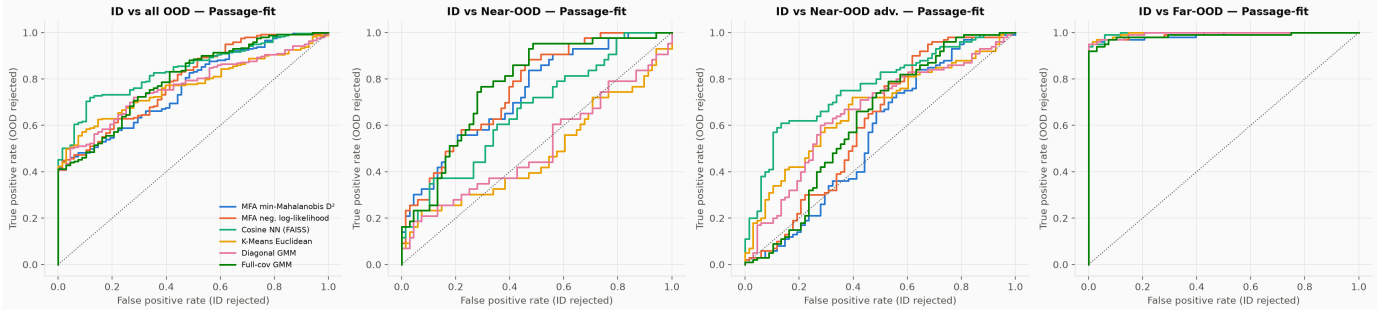


Figure 5. ROC curves per OOD type: the closer to the top-left, the better; the diagonal is chance.

- **Far-OOD is solved** by every method (AUROC ≥ 0.986).
- **On real Near-OOD questions** (companies not in the knowledge base), covariance-aware densities win. A missing company shifts the embedding in a *direction*, not just a distance, which spherical K-Means and the axis-aligned

diagonal GMM cannot see (≈ 0.5). The MFA beats cosine similarity by +0.09 AUROC (log-likelihood; bootstrap 95% CI [−0.004, 0.19], MFA ahead in 97% of resamples).

- **False premises are invisible to density.** Adversarial questions use indexed companies' own vocabulary, so they look in-distribution (MFA 0.56). Cosine does better (MFA − cosine = −0.20, CI [−0.30, −0.11]) because invented entities have no close chunk.
- **Full covariance is affordable at this scale** (~110 training points per dimension), so the full GMM is slightly ahead of the MFA on Near-OOD (0.772 vs 0.756) while the MFA uses 69% fewer parameters and fits 3× faster.

4.2 Operating points

Rule	ID pass ↑	Near-OOD rejected	Adversarial rejected	Far-OOD rejected
MFA + EVT, α = 0.01 (proposal)	**100%**	4.7%	0%	84%
MFA + EVT, α = 0.1	92.6%	**32.6%**	5%	**97%**
MFA + EVT, question-calibrated α = 0.1 (3-fold CV)	92.6%	30.2%	4%	97%
Cosine NN, 99th validation percentile	72.1%	44.2%	64%	100%
Full-cov GMM, 99th validation percentile	76.5%	58.1%	29%	98%

α = 0.01 never refuses a genuine question but lets nearly all Near-OOD through. **α = 0.1 is the better trade-off.** Calibrating on ID questions instead of passages gives the same point, so passage calibration transfers on this data.

4.3 Downstream hallucination

Each question is answered once by FAISS top-3 retrieval + Qwen2.5-1.5B-Instruct (greedy). The guarded conditions replace blocked questions' answers with a fixed refusal, so every difference comes from the gate. Two prompts were used: *naive* ("use the context to answer") and *strict* ("answer only from the context, otherwise say you don't know"). **Hallucination rate** is the share of OOD questions that get a substantive answer; **faithfulness** is the share of answer sentences entailed by the retrieved chunks, judged by an NLI model.

	Unguarded	Guarded α = 0.01	Guarded α = 0.1 (q-calibrated)
Hallucination, all OOD, naive prompt	44.4%	**20.2%**	**14.8%**
Near-OOD / adversarial / Far-OOD, naive	72% / 10% / 67%	72% / 10% / 8%	58% / 9% / 2%
Hallucination, all OOD, strict prompt	11.5%	9.5%	7.4%
ID questions wrongly blocked	0%	**0%**	7.4%
ID faithfulness (answered)	0.43	0.43	0.43

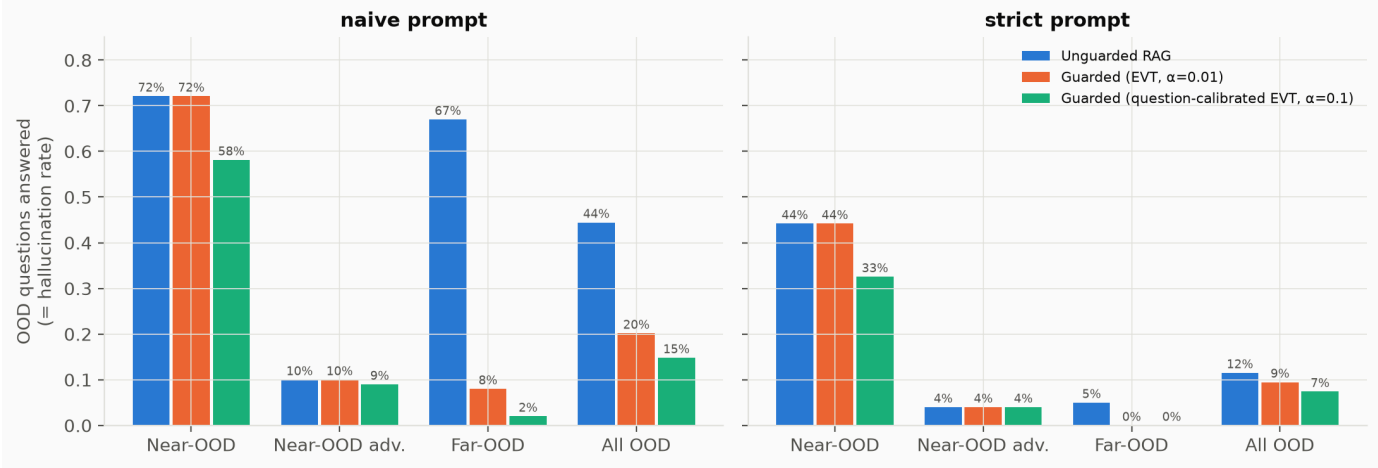


Figure 6. Share of OOD questions answered (= hallucinated) without and with the guardrail.

With the naive prompt, the guardrail **halves OOD hallucinations at no cost to ID questions** (α = 0.01) and removes two-thirds at α = 0.1, mainly by stopping trivia questions. Near-OOD questions remain the main risk: even with the strict prompt, the LLM answers 44% of them from *other* companies' filings, and the guardrail lowers this to 33%. 35–47% of OOD questions never reach the LLM, which saves their generation cost.

4.4 Latency

Over 1,000 questions on CPU the guardrail takes **0.77 ms (p50) / 1.45 ms (p95)**, under the 10 ms target: ~2,000× faster than inverting Σ_k per question (2.9 s) and less than the MiniLM embedding (7.7 ms) that RAG already pays.

5. Discussion and limitations

- **What worked:** Far-OOD detection; a covariance-aware density for real Near-OOD questions; a well-fitting, stable EVT threshold; and a ~1 ms guardrail that halves hallucinations without blocking genuine questions.
- **What did not:** false-premise questions about indexed companies cannot be detected from the question's embedding; they need post-retrieval answerability or NLI checks. At this data size, full-covariance models are as good as the MFA, whose savings matter for larger d or less data per cluster.
- **Pilot vs final:** an earlier pilot with 7 big-tech filings and generated questions (branch `old`) found cosine ahead of the MFA (0.894 vs 0.858). Its Near-OOD set consisted only of false-premise questions; with real FinanceBench questions the ranking reverses.
- **Limitations:** small question sets (68 ID, 43 Near-OOD; AUROC CIs ≈ ±0.09); FinanceBench is hard for a 1.5B model with top-3 MiniLM retrieval (the right filing is retrieved for 39 of 68 ID questions); one greedy answer per question, generated in fp16 on a GPU; NLI judging is strict on paraphrased numbers; the selected rank lies on the grid edge.
- **Next steps:** score by MFA log-likelihood, combine the density gate with cosine similarity, add a post-retrieval answerability check, and test 768-d embeddings.

6. Related work

- Lee et al. (2018), *A Simple Unified Framework for Detecting OOD Samples and Adversarial Attacks*, NeurIPS: Gaussian class-conditional densities with Mahalanobis scores. This project replaces the shared covariance with per-cluster low-rank covariances.
- Ren et al. (2023), *Out-of-Distribution Detection and Selective Generation for Conditional Language Models*, ICLR: Mahalanobis-distance OOD detection on language-model embeddings to decide when to abstain, the closest analogue to this guardrail.

7. Data and tool references

SEC EDGAR and its APIs · FinanceBench (CC-BY-NC-4.0) · Open Trivia DB · all-MiniLM-L6-v2 · Qwen2.5-1.5B-Instruct · nli-deberta-v3-base · FAISS · RAG, Lewis et al. (2020). Reproduction steps: README.